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Wake Deflection Control with Wind Direction Changes: Wind Tunnel Comparison of Different Wind Farm Flow Models

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Abstract— **The paper studies the effects of model fidelity on the performance of a robust open-loop wake deflection controller. For this purpose, a small cluster of scaled turbines is operated in a boundary layer wind tunnel, where wind direction changes are simulated by a rotating turntable. Results indicate that better models lead to improved power gains and reduced fatigue loading, motivating the development of improved accuracy wind farm flow models.**

I. Introduction

In recent years, there has been a growing interest in the development of cooperative control strategies for wind farms. Using wake steering, which appears to be the most promising approach [1], [2], wind turbines are intentionally misaligned with respect to the ambient wind direction. A simple but effective way to implement such control logic is to compute off-line look-up tables (LUTs), using a suitable wind farm flow model. The LUTs are then used at run time to get the optimal yaw misalignment of each wind turbine according to the current ambient wind conditions. This approach seems also promising for practical field deployment. In fact, wake steering field tests have already been reported in the literature [3], [4], and a control technology based on wake steering is offered as a product by one of the leading wind turbine manufacturers [5].

Although a field demonstration is clearly the final litmus test of any wind technology, simulations with high fidelity models and scaled testing in wind tunnels offer some unique opportunities to improve knowledge and understanding. In this regard, Ref. [6] described some pioneering wind tunnel experiments performed with actively controlled and extensively sensorized scaled wind turbines. Building on that initial experience, an experimental demonstration of closed-loop wake steering control was later presented in [7], which also generated a valuable data-set used for the validation of simulation models [8].

The present paper follows in these same tracks and tries to answer the following scientific question: how accurate should a wind farm flow model be for effective open-loop wind farm control? In other words, is it necessary to use a high accuracy model, or is it sufficient to use a fairly rough and simple model? This question is particular relevant for open-loop control, as there is no feedback in this case that can help mitigate the possible deficiencies of the underlying model used for control synthesis. Wind tunnels offer an opportunity to try to —at least in part— answer these questions, given their known, controllable and repeatable conditions.

Wind tunnel tests were conducted with three scaled wind turbines. To make the tests even more realistic, dynamic wind direction changes were simulated by rotating the wind tunnel turntable. Control LUTs were computed using three different wind farm flow models: an implementation of the FLORIS model [9] tuned using turbine wake measurements [10], a data-augmented version of FLORIS (termed FLORIS-Augm) that captures inflow non-uniformity and secondary steering [11], and a very high-accuracy purely Data-Driven model [12].

Results indicate that the choice of a flow model does make a difference on the performance of the controllers, notably in power capture and loading.

II. Experimental Setup

The experimental setup is reported in Fig. 1: a cluster of three scaled wind turbines is installed on the 13 m diameter turntable of the atmospheric test section of the Politecnico di Milano wind tunnel [6]. The turntable can be rotated to simulate different wind directions by a dedicated controller that tracks a desired rotation time history using, as feedback, the signal provided by an optical encoder. The wind direction Φ is null when the imaginary line that passes through the three rotor hubs is parallel to the wind tunnel center-line, and positive for a clockwise rotation of the turntable when viewed from the top, as shown in Fig. 1. The wind turbines are longitudinally spaced of 5 diameters (D) and are located 1.5D to the left of the turntable center-line, when looking upstream.

An atmospheric boundary layer typical of offshore wind condition was obtained by the use of spires placed at the inlet of the test section. A vertical scan of the generated inflow was obtained 4D upwind of the most upstream turbine (labelled WT1) with a three-component CTA hot-wire probe. The measured vertical wind profile is best-fitted by an exponential law with exponent $\alpha = 0.144$, while the turbulence intensity at hub height is approximatively equal to 6%.

A pitot tube is installed at hub height 3D upwind of WT1, outside of the turntable, to provide the reference ambient wind speed. The flow generated in the wind tunnel is not exactly uniform in the lateral (horizontal cross-flow) direction [13]. Hence, when the turntable is rotated, the

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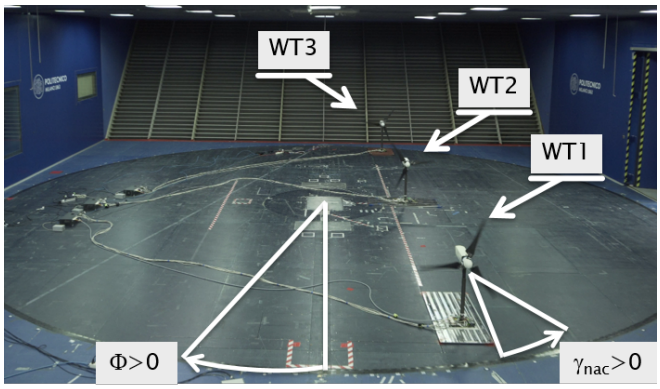


Fig. 1: Wind tunnel setup for the variable wind direction wind farm control experiment

turbines are exposed to a local ambient flow that is not exactly equal to the one measured by the pitot tube. By computing the local rotor equivalent wind speed at each turbine location with the torque balance equation [14], these differences were found to be of up to $\pm$ 4%, depending on the relative position between pitot tube and rotor.

A. Wind Turbine Model

The wind turbines used in this work are three identical G1 scaled models, whose rotor diameter, hub height and rated rotor speed are 1.1 m, 0.825 m and 850 rpm, respectively. The models, already used in several previous research projects [7], [15], [16], are actuated in pitch, torque and yaw, and are equipped with load sensors at the shaft and tower base. Further details about the G1 design, its aerodynamic performance and several applications can be found in [17].

Each G1 is controlled by its own real-time Bachmann M1 system, where supervisory control functions, pitch-torque-yaw control algorithms and all necessary safety, calibration and data logging functions are implemented. A standard power controller is implemented based on [18]. The nacelle orientation γ_{nac}, positive for a counter-clockwise rotation viewed from the top (as shown in Fig. 1), can be varied at will with respect to the ground. By knowing the local wind direction, one can thus achieve a desired misalignment angle γ with respect to the incoming wind vector. A positive γ corresponds to a counter-clockwise rotation from the wind to the rotor axis looking down onto the terrain. With these definitions, the relationship among wind direction, nacelle orientation and yaw misalignment angle is

$$\gamma = \gamma_{\mathrm{nac}} - \Phi. \tag{1}$$

Using the present control implementation, the rotor thrust (C_{T}) and power (C_{P}) coefficients vary as shown in Fig. 2. The relationship between the rotor effective wind speed U_{REWS} and C_T-C_P is reported in the two upper plots, while the experimentally measured relationship between γ and C_T-C_P is reported in the lower plots. The figures also report the cosine-laws that best-fit the experimental data, the power and thrust loss exponents being equal to 2.174 and 1.425, respectively.

B. Dynamic Wind Direction Changes

In this work, open loop wake deflection is tested in the presence of dynamically varying wind direction conditions. For this purpose, the time speed-up induced by the scaling laws should be taken into consideration. This speed-up, defined as the ratio n_Ω between the rated rotor speed of the scaled model and of its full-scale reference [6], is used to time-scale the wind direction time history. In this specific case, the full-scale reference is a generic wind turbine model with a rated power of 8 MW and a rated rotor speed of 10.5 rpm [19], resulting in a time speed-up $n_\Omega = 80$.

In order to obtain significant results that could be up-scaled, wind direction changes should be realistic, i.e. they should mimic changes observable at full-scale. For this purpose, 5 days of met-mast measurements of the wind direction were considered, using 1 Hz data reported in [20]. A comparison between the spectra of the wind directions measured in the field with the met-mast (Φ_{met}) and in the wind tunnel with the CTA probes revealed that the latter lack low-frequency and large-amplitude fluctuations [12]. In the experiment, these fluctuations were therefore reproduced with the wind tunnel turntable. Specifically, a time history of the rotation of the turntable was defined to reproduce the scaled 2-min average of the met-mast data taking into account, however, limitations imposed by the maximum speed of actuation of the turntable. The resulting 90-min time history of turntable rotations Φ_{turn} is compared to the scaled 2-min average of the met-mast data in Fig. 3.

III. Open-Loop Wake Deflection Controller

The wind farm control strategy employed here is the open-loop algorithm sketched in Fig. 4. Similar control strategies have also been recently implemented in the field [4].

The controller receives as input ambient wind conditions, including wind direction, wind speed and turbulence intensity. Filtering is used to remove fast fluctuations, obtaining a control behavior that only reacts to low-frequency changes [4], [21]. Based on the input quantities, the controller interpolates within a LUT to obtain desired yaw misalignments with respect to the wind direction of each individual wind turbine, which are in turn dispatched to the respective turbines. For each machine, the yaw misalignment is applied as an offset to the current set point of the turbine yaw controller. In this sense, the original yaw control logic of the turbine is unaffected, which eases the implementation of wake deflection on existing turbines.

A. Controller Implementation

The wind farm controller was implemented on a desktop PC, and the MODBUS protocol was used for the communication between the PC and the Bachmann modules of each wind turbine. The control step was set to 0.75 sec, which corresponds to 1 min at full-scale.

Average wind speed and turbulence intensity were kept constant in the present experiments, while they would clearly vary in a field test [4]. Therefore, only the wind direction was used as input to the LUT. The wind direction was obtained

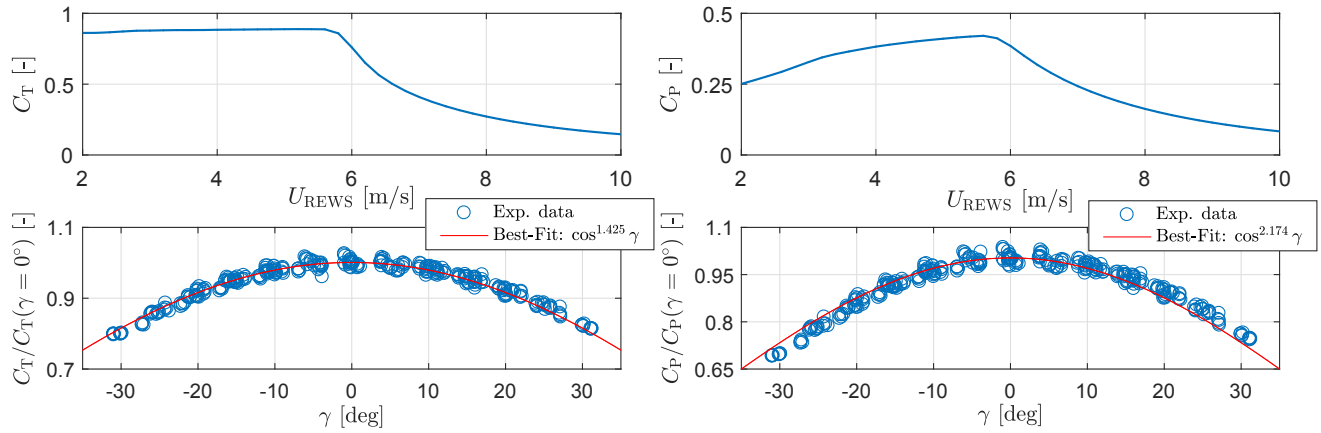


Fig. 2: G1 model regulation: thrust (upper left plot) and power (upper right plot) coefficients vs. rotor equivalent wind speed; thrust (lower left plot) and power (lower right plot) coefficients vs. yaw misalignment

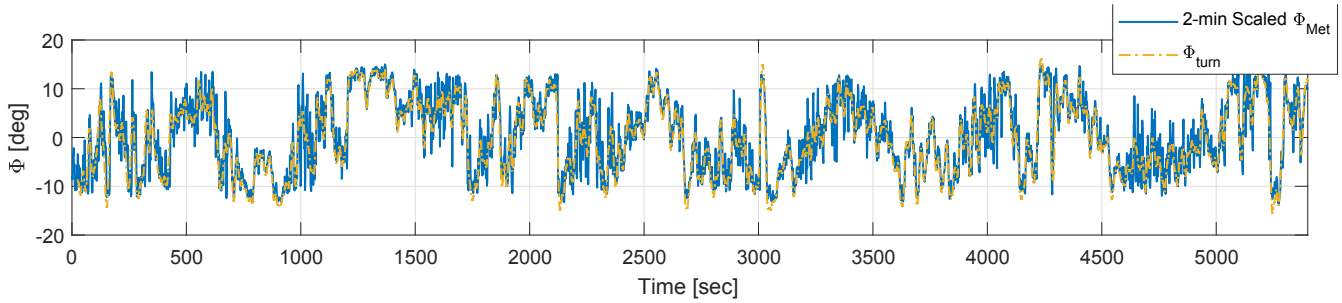


Fig. 3: Comparison between the turntable rotation Φ_{turn} and the scaled 2-min average of the met-mast wind direction Φ_{met}

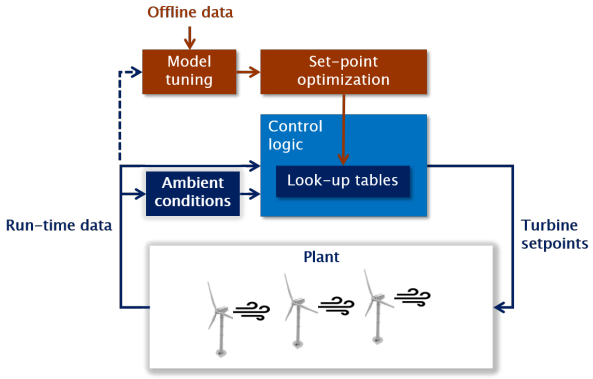


Fig. 4: Wind farm control scheme. The dashed line (model updating based on run-time data) is not used in the present work

by the measured turntable rotation Φ_{turn}, while it would have to be estimated (for example, from the wind turbine yaw attitude and/or met masts) in a field implementation. A moving average with a length of the sliding window equal to $T_{\text{MAvg}} = 7.5$ sec was used to filter the wind direction, which corresponds to 10 min at full-scale. This filtering action implies an uncertainty in the wind direction, which can be quantified by the standard deviation σ_Φ of the difference Δ_Φ between the unfiltered and filtered signals. The histogram of Δ_Φ is shown in Fig. 5, together with its fitted Gaussian normal distribution.

The LUTs were pre-computed off-line for all relevant expected ambient conditions using three different wind farm flow models, described in the following. Yaw offsets were applied only to the two upstream wind turbines, which were actuated with a maximum yaw rate of $\dot{\gamma}_{\max} = 10$ deg/s (corresponding to 0.125 deg/sec at full scale) to limit gyroscopic loads on the G1. The value chosen here is lower than the 0.3 deg/sec yaw rates used in other publications [22], [23]. Yaw rate has indeed a significant effect on performance, but also on loads, duty cycle and on the overall design and sizing of the yaw system, so that its optimal choice requires a system-level analysis that is beyond the scope of this work. For a given measured wind direction Φ_{meas} and desired yaw misalignment γ, the nacelle orientation was computed from (1) as $\gamma_{\text{nac}} = \gamma + \Phi_{\text{meas}}$. The direct yaw control method used in this work differs from the indirect one of [3], [4] and was selected because it provides for a more precise yaw offset tracking [24].

B. Wind Farm Models

The wind farm models used for the synthesis of the LUTs include an analytical engineering model, a purely data-driven one, and an hybrid between the two.

The calibration and generation of these models was obtained by a subset of extensive wind tunnel time-averaged

measurements, including: wake profiles in various environmental and operating conditions; power outputs of the three turbines, for 11 different mean wind directions ($\Phi = \pm$ 11.54, $\pm$9.21, $\pm$6.89, $\pm$4.58, $\pm$2.29 and 0) and for a broad range of yaw misalignments of the two upstream wind turbines. To account for the lateral non-uniformity of wind tunnel inflow, the power coefficient of the jth machine was computed as $C_{\mathrm{P},j} = P_j/1/2\rho A U_j^3$, where U_j is the wind speed measured by the reference pitot transducer multiplied by a factor that accounts for the wind velocity at that location.

The formulation of the three models is briefly described in the following sections, including the relevant references to which the reader is referred to for additional details.

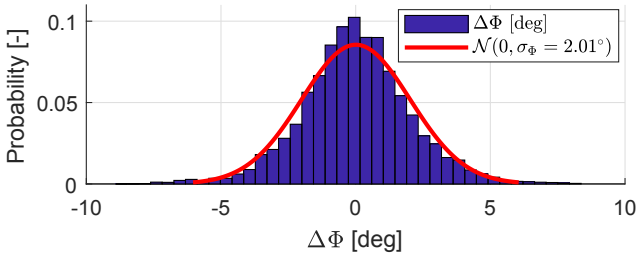


Fig. 5: Probability distribution of the difference Δ_Φ between the turntable position and its filtered value

1) FLORIS Model: the FLORIS implementation used here is based on the online version [9], which incorporates a plethora of sub-models. Specifically, here the *selfSimilar* velocity deficit, the *rans* deflection and the wake model by [25] are used. Finally, *quadraticRotorVelocity* and *crespoHernandez* [26] have been selected as wake combination and added turbulence sub-models, respectively.

The parameters of the FLORIS model were identified based on measurements of the wake of a single G1 turbine, as discussed in [10]. The $C_{\mathrm{T}}(U)$ and $C_{\mathrm{P}}(U)$ curves are the ones depicted in the upper part of Fig. 2, while power losses due to yaw misalignment are modeled with the cosine-laws of the lower plots of the same figure. The ambient wind field within the FLORIS model is based on measurements of the wind tunnel inflow, and it is therefore laterally not-uniform. Finally, power is computed by integrating the flow at the rotor disc as $P = 1/2\rho \int_A V^3 C_{\mathrm{P}} \, \mathrm{d}A$.

2) Data-Augmented FLORIS Model: an augmented version of the baseline FLORIS model (termed FLORIS-Augm in the following) was developed following the approach described in [11]. Specifically, the standard FLORIS model was augmented with error terms that account for non-uniform inflow and secondary steering. This latter effect results in changes in the local ambient wind direction at a downstream rotor caused by an impinging wake shed by a yaw-misaligned upstream turbine [27]. The error terms are modeled with parametric functions, "surgically" inserted into the model equations to represent expected deficits of the model. The error terms are then identified (or learnt) from operational data (power, in this case). Since similar data is typically available from standard SCADA (Supervised Control and Data Acquisition) systems, this approach can also be used at full-scale in the field, as discussed in [11].

3) Data-Driven Model: A model purely based on experimental data was obtained by best-fitting with shape functions measured $C_{\mathrm{P},j}$ values, this way obtaining a response surface. The shape functions were inspired by experimental observations and the modelling of wake superposition, as described in more detail in [12]. Due to the fact that it is based on a large dataset of accurate wind tunnel measurements, the Data-Driven model provides for a very precise representation of the power output of the turbines. However, on the down side, one should note that this same approach would hardly be usable in the field.

Model Comparison: The plots in Fig. 6 report, for Φ = 0 deg, the normalized measured power of the wind farm, defined as $C_{\mathrm{P,WF}} = C_{\mathrm{P},1} + C_{\mathrm{P},2} + C_{P,3}$, together with the predictions of the baseline FLORIS (top), FLORIS-Augm (center) and Data-Driven (bottom) models. The resulting root mean square errors e_{RMS}, expressed in percent of the available upstream wind power and included in the legends, allow one to appreciate the increasing level of fidelity to reality of these models. Similar conclusions are obtained for different wind directions, as reported in [11].

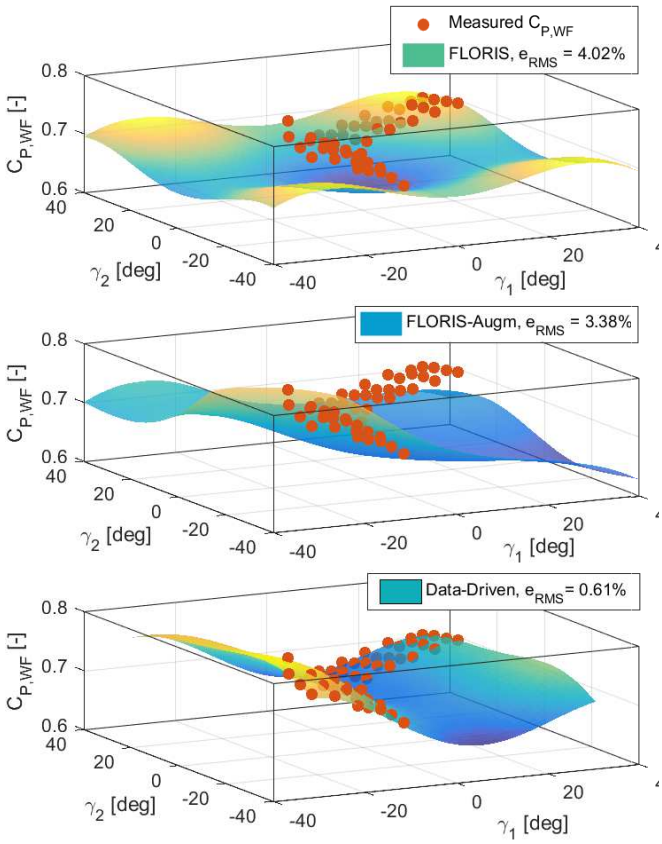


Fig. 6: Measured wind farm power coefficient $C_{\mathrm{P,WF}}$ compared to the predictions of the baseline FLORIS (top), FLORIS-Augm (center) and Data-Driven (bottom) models for $\Phi = 0$ deg

C. Look-Up Table Computation

The control LUTs were derived for a set of wind directions Φ characterized by a resolution of 0.2 deg for $\Phi \in [-2, 2]$ deg, and a resolution of 1 deg outside of this range. For the baseline FLORIS and FLORIS-Augm models, a constant below-rated mean wind speed equal to 5.25 m/s was assumed, which is the mean wind speed measured by the reference pitot tube during the wind tunnel tests. Recent research [21], [28], [29] discusses the importance of including wind direction and yaw uncertainties in the calculation of the control LUTs. In this work, only uncertainties in wind direction are taken into account as they were expected to be most significant. Yaw uncertainties due to the limited yaw rate and possible sensor errors are therefore not taken into account. The approach proposed and discussed in [29] was used to obtain, for each model, different robust LUTs corresponding to the four uncertainty levels $\sigma_\Phi = \{0, 2, 4, 6\}$ deg.

The optimal yaw misalignments γ_1^* (for WT1) and γ_2^* (for WT2), limited to the range between ± 30 deg, are computed as

$$[\gamma_1^*, \gamma_2^*] = \arg\max_{\gamma_1,\gamma_2} \sum_{k=0}^{9} P_{\mathrm{M}}\left(\Phi + \Delta\Phi_k, \gamma_1 - \Delta\Phi_k, \gamma_2 - \Delta\Phi_k, -\Delta\Phi_k\right) f(\Delta\Phi_k), \quad (2)$$

where $\Delta\Phi_k = (4k/9 - 2)\,\sigma_\Phi$ is the uncertainty associated with the wind direction measurement varying in the range of $\pm 2\sigma_\Phi$, $f \sim \mathcal{N}(0, \sigma_\Phi)$ is the Gaussian normal distribution and $P_{\mathrm{M}}(\Phi, \gamma_1, \gamma_2, \gamma_3)$ the wind farm power predicted by the flow model. This optimization problem is solved using the pattern-search algorithm, as implemented in Matlab.

The resulting LUTs for $\sigma_\Phi = 0$ deg are shown in Fig. 7. It is interesting to note that, for the Data-Driven model, the misalignment angles for the second turbine are smaller than the ones of the first one, especially for positive wind directions. This is in agreement with other recent studies [7], [30].

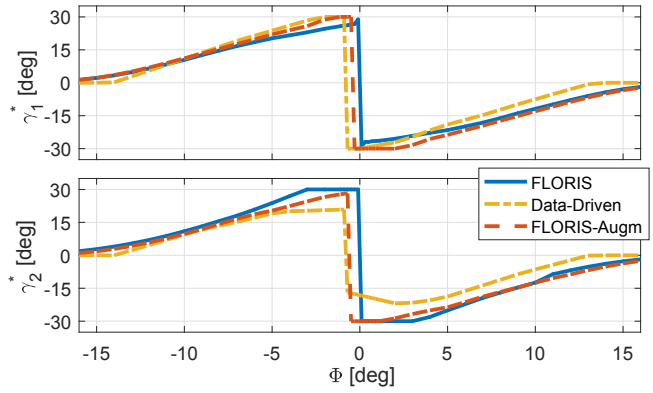


Fig. 7: Optimal yaw misalignments for WT1 (top) and WT2 (bottom) computed with the baseline FLORIS, FLORIS-Augm and Data-Driven models for $\sigma_\Phi = 0$ deg

IV. Results

A. Expected Gains with Constant Mean Wind Directions

As a first step, the wind farm power output was determined in constant mean wind directions with the greedy control policy (i.e., no wake steering) and with the LUT-based controllers in the absence of wind uncertainties ($\sigma_\Phi = 0$ deg). As plant, the Data-Driven model was used for this purpose, being essentially an exact representation of the wind farm behavior, except for measurement errors. Figure 8 reports the expected pitot-normalized power gain $\Delta C_{\mathrm{P,WF}}$, evaluated with respect to the greedy case, for $\Phi = \pm$ 11.54, $\pm$9.21, $\pm$6.89, $\pm$4.58, $\pm$2.29 and 0 deg.

The plot indicates that all different models lead to positive gains for each wind direction, up to about 25% in the best conditions. The gains obtained with the baseline FLORIS model are only slightly lower than those of the FLORIS-Augm and the Data-Driven models. This appears to indicate that the cost function of problem (2) is rather insensitive to the details of the underlying model when no uncertainties are considered. Considering the wind direction time series shown in Fig. 3, under the assumption of an exact knowledge of the wind direction and an instantaneous realization of the required yaw offset, power gains equal to 10.73%, 11.52% and 11.84% are expected if, respectively, the FLORIS, FLORIS-Augm and Data-Driven LUTs are used.

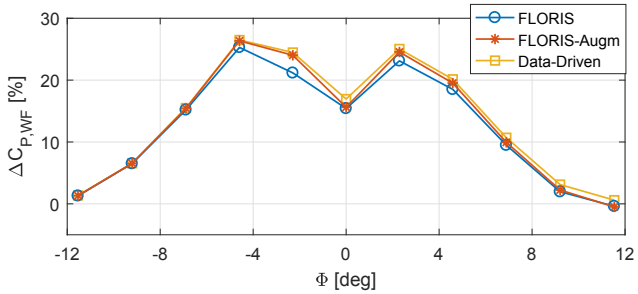


Fig. 8: Experimental gain $\Delta C_{\mathrm{P,WF}}$ in wind farm power output vs. wind directions Φ, for the different models and in the $\sigma_\Phi = 0$ case

B. Results with Dynamic Changes of Wind Direction

Next, wind tunnel tests were conducted in variable wind direction conditions, by using the turntable. The prescribed direction time history was repeated multiple times: each test was performed for a total of 1.5 h divided in 9 intervals of 600 sec, in order to allow for the periodic re-calibration of the wind tunnel and of the various turbine sensors. The mean wind speed measured by the reference pitot tube was approximately equal to 5.25 m/s.

1) Gains in Wind Farm Power Output: Figure 9 shows the obtained normalized power gains with respect to the greedy case.

Results indicate power increases of about 4-6%, a range that is considerably lower and broader than the one previously estimated in section IV-A. Most probably, the lower

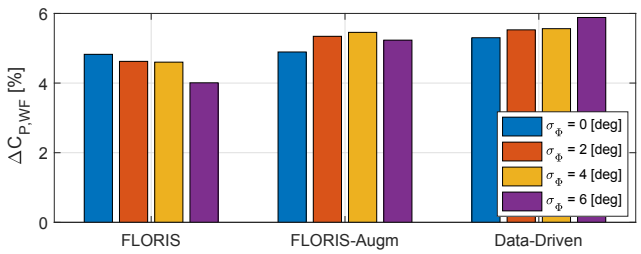


Fig. 9: Overall gain $\Delta C_{\mathrm{P,WF}}$ in wind farm power output with respect to the greedy case for the three models and various uncertainty levels σ_Φ

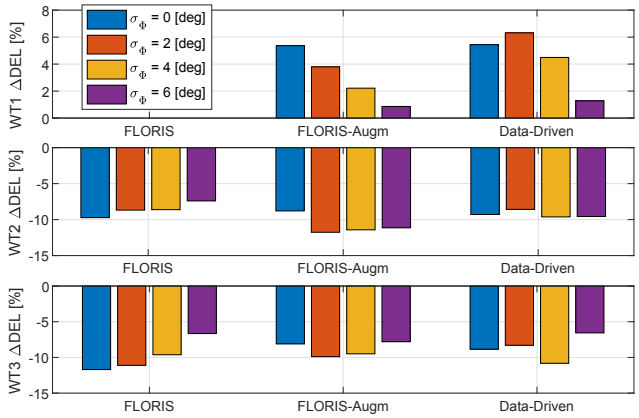


Fig. 10: Change in rotating hub DELs with respect to the greedy case for the three models and various uncertainty levels σ_Φ

power increase can be attributed to wind direction uncertainties, limited turbine yaw rate and wake propagation dynamics. The best performance is obtained with the Data-Driven model and for $\sigma_\Phi = 6$ deg. Additionally, the results of the FLORIS-Augm model are better than the ones of the baseline FLORIS. For both the Data-Driven and the FLORIS-Augm models, robust LUTs exhibit higher gains than LUTs that neglect wind direction uncertainties ($\sigma_\Phi = 0$ deg). This is not the case for the baseline FLORIS, which shows the opposite behavior.

These results allow to make some interesting considerations. First, if the model is strongly biased, as in the present FLORIS case, introducing robustness in the formulation may decrease performance, which is in contrast with the results reported by [29]. On the contrary, robustness increases performance if the underlying models have better fidelity, which is the case here for the FLORIS-Augm and Data-Driven models. Even more importantly, better models lead to better performance. In addition, one should also note that maximum gains are obtained for $\sigma_\Phi = 4$ and 6 deg, respectively for the FLORIS-Augm and Data-Driven models. These uncertainty values are significantly higher than the ones estimated earlier due to filtering and reported in Fig. 5. This aspect requires further analysis.

2) Impact on Loads and Actuator Duty Cycle: after evaluating the impact on power, the effects of the various models on loads and on the yaw duty cycle were evaluated. Figure 10 shows the variation of the hub rotating damage equivalent loads (DELs), with respect to the greedy control policy. The loads for WT1 in the baseline FLORIS case are not reported in the figure, due to a problem with the recording of loads during these tests.

A few observations can be made by looking at these results. First, by pointing the rotor away from the wind, the DELs of the front machine increase, as expected. However, it is particularly interesting to note that this increase is marginal when robust LUTs computed with $\sigma_\Phi = 6$ deg are used, which still correspond to significant power gains as shown in Fig. 9. Second, wake steering control is particularly beneficial if one looks at the DELs of the second and third machines; their reduction, indeed, varies between 7-12%, depending on the LUTs. Interestingly, with LUTs computed with the baseline FLORIS and FLORIS-Augm models, DEL reductions seem to be well correlated with power gains: the robust LUTs with the largest power gains also generate the maximal loads reductions. On the other hand, the results for the Data-Driven LUTs do not present a clear trend.

To evaluate the impact of wake steering on yaw control effort, the wind farm yaw actuator duty cycle is defined as

$$\mathrm{ADC_{WF}} = \frac{1}{3}\sum_{j=1}^{3}\frac{1}{T}\int_0^T \frac{|\dot{\gamma}_{\mathrm{nac},j}(t)|}{\dot{\gamma}_{\max}}\,\mathrm{d}t, \tag{3}$$

where $\gamma_{\mathrm{nac},j}(t)$ is the time history of the nacelle orientation of the j-th wind turbine. The wind farm actuator duty cycle is an indicator of the average usage of the yaw actuators, and could therefore be used to quantify the impact of wake steering control on the maintenance cost of the yaw drives at the wind farm level. Figure 11 shows the increase $\Delta\mathrm{ADC_{WF}}$ of the wind farm yaw actuator duty cycle with respect to the greedy control strategy. As expected, the use of wake steering control has a significant effect on $\mathrm{ADC_{WF}}$, which can be in part mitigated by the use of robust LUTs with increasing wind direction uncertainty. On the other hand, the model used for the synthesis of the LUTs seems to have only a minor effect on this performance indicator.

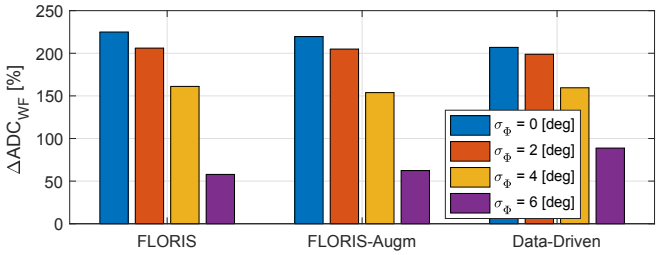


Fig. 11: Change in wind farm yaw duty cycle with respect to the greedy case for the three models and various uncertainty levels σ_Φ

V. Conclusions

This study has analyzed the effects of wind farm flow model fidelity on the performance of open-loop wake steering control. Furthermore, the study has considered the effects caused by wind direction variations. Based on the results reported herein, the following conclusions can be drawn:

- Higher model accuracy leads to improved power capture and reduced loading, while it has a negligible effect on control effort. From this point of view, it is worthwhile to improve wind farm models.
- Model augmentation is an effective way to increase the fidelity of a baseline engineering model.
- The use of a robust approach is of crucial importance for performance and robustness.
- As a final marginal note, it should be reported that there is an apparent discrepancy in the optimal value of the uncertainty observed in the present tests, given that the true uncertainty is known in this case. This aspect necessitates of further investigations in order to be clarified.

Future work will aim at developing and testing dynamic closed-loop controllers, in order to understand and quantify their potential benefits with respect to the simple steady-state open-loop approach describe in this paper. Additionally, more sophisticated cost functions will be investigated that are able to trade between the different requirements of power capture and fatigue damage.

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